

FINAL PROJECT PROPOSAL

SecurAI Mobile

The AI-Powered API Security Copilot

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Track 7 (API/Application Security) x Track 9 (AI Automation)
THE ARZENS Advanced Track Engineering Internship Program

Project Title

SecurAI Mobile: The AI-Powered API Security Copilot

Problem Statement

Security engineers and developers often need to monitor API health, respond to security alerts, and review code vulnerabilities while away from their desks. Existing security platforms are heavily desktop-focused, leading to slower response times during critical incidents. Furthermore, manual triage of API vulnerabilities and alert fatigue overwhelm security teams. There is a strong need for an intelligent, mobile-first solution that automates triage and provides on-the-go security assistance.

Proposed Solution

SecurAI Mobile is a cross-platform mobile application that serves as a Security Copilot for development and security teams. Combining API security monitoring (Track 7) with AI automation (Track 9), the app connects to backend API gateways to monitor traffic and vulnerabilities. It utilizes a Private RAG (Retrieval-Augmented Generation) system to provide an AI assistant capable of answering security queries, automatically generating incident reports, and suggesting secure coding fixes directly from a mobile device.

Project Objectives

- Develop a mobile interface for real-time monitoring of API authentication and security metrics.
- Integrate an AI chatbot (Copilot) that uses Private RAG to query internal security documentation and vulnerability reports.
- Automate the generation and distribution of security incident reports via the mobile app.
- Provide actionable, AI-generated secure coding recommendations for detected API flaws.

Key Features

- **Real-time API Security Dashboard:** Visualizing API traffic, authentication failures, and active threats on a mobile interface.

- **AI Security Copilot Chat:** An intelligent assistant where users can ask questions like, “What are the latest vulnerabilities in the payment API?” or “How do I fix a broken object-level authorization (BOLA) flaw?”
- **Automated Alerting & Reporting:** Push notifications for critical API anomalies and one-click AI-generated incident reports (PDF/Email).
- **Private RAG Knowledge Base:** The AI is grounded in the specific organization's API documentation and security policies, ensuring relevant and accurate advice instead of generic AI answers.

Technologies & Tools to be Used

- **Frontend (Mobile):** Flutter or React Native (for cross-platform iOS/Android development).
- **Backend:** Node.js (Express) or Python (FastAPI).
- **AI/LLM Integration:** OpenAI API (GPT-4o) or Gemini API.
- **RAG Implementation:** LangChain and a Vector Database (like Pinecone, ChromaDB, or Firebase Vector Search) for storing and retrieving security documents.
- **Security Integration:** Mock integration with an API Gateway or a vulnerability scanner (e.g., OWASP ZAP) via webhooks to simulate incoming security data.

Expected Outcome

A functional mobile application demonstrating how AI can streamline API security management. It will showcase a working AI Copilot that can query a private knowledge base, monitor simulated API traffic, and generate automated security reports, proving the viability of mobile DevSecOps tools.

Challenges You May Face

- **Implementing Private RAG:** Effectively chunking and embedding security documentation so the AI retrieves accurate context without hallucinating.
- **Mobile UI/UX Design:** Designing a clean, intuitive interface that makes complex security data readable on a small mobile screen.
- **Simulating Realistic Data:** Creating realistic mock API logs and vulnerability data for the app to monitor and analyze during the demonstration.

Future Improvements

- Integration with Jira/Slack to automatically create tickets from the mobile app.
- Direct integration with GitHub/GitLab to approve AI-generated code fixes (Pull Requests) directly from the phone.
- Voice-command capabilities for hands-free security querying.

Why I Chose This Project

This project perfectly bridges Track 7 and Track 9 by taking traditional application and API security concepts and supercharging them with modern AI automation. It addresses a real-world problem, alert fatigue and the lack of mobile DevSecOps tools, while allowing me to showcase practical skills in mobile app development, backend architecture, and cutting-edge generative AI (RAG) implementation. Building a Copilot demonstrates a forward-thinking engineering mindset.